

| Section | I 1-2 | I 3 | I 4-5 | Q6  | Q7 a) | Q7 b) | Q7 c) | Total |
|---------|-------|-----|-------|-----|-------|-------|-------|-------|
| E3      | /2    |     | /2    |     | /7    |       |       | /11   |
| E4      |       |     |       |     |       | /6    |       | /6    |
| E8      |       | /1  |       | /16 |       |       | /7    | /24   |
|         |       |     |       |     |       |       |       | /41   |

## Section I: Multiple Choices

5 marks

Attempt all questions.

Answer on the multiple choice answer sheet provided for Section I.

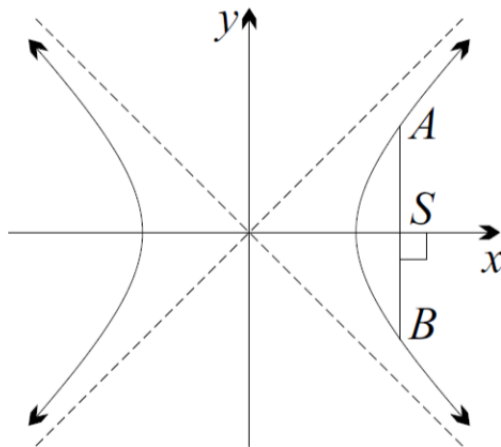
---

1. What type of curve does the equation  $|z - 3| - |z + 3| = 4$  corresponds to in the Argand diagram?  
  
(A) Circle  
(B) Ellipse  
(C) Parabola  
(D) Hyperbola
  
2. The point  $T(a \cos \theta, a \sin \theta)$  lies on the circle  $x^2 + y^2 = a^2$ . Which of the following is the equation of the tangent at  $T$ ?  
  
(A)  $x \cos \theta + y \sin \theta = a$   
(B)  $x \cos \theta + y \sin \theta = a^2$   
(C)  $x \cos \theta - y \sin \theta = a$   
(D)  $x \cos \theta - y \sin \theta = a^2$
  
3. Given that  $\sin(A - B)x + \sin(A + B)x = 2 \sin Ax \cos Bx$ , which of the following is equal to

$$\int \sin 3x \cos 2x \, dx ?$$

- (A)  $\frac{1}{2} \sin 5x + \frac{1}{2} \sin x + C$
- (B)  $\frac{1}{10} \cos 5x + \frac{1}{2} \cos x + C$
- (C)  $-\frac{1}{2} \sin 5x - \frac{1}{2} \sin x + C$
- (D)  $-\frac{1}{10} \cos 5x - \frac{1}{2} \cos x + C$

4. Which of the following are the foci of the ellipse  $\frac{x^2}{9} + \frac{y^2}{4} = 1$  ?
- (A)  $(2,0)$  and  $(-2,0)$
- (B)  $(3\sqrt{5},0)$  and  $(-3\sqrt{5},0)$
- (D)  $\left(\frac{3\sqrt{5}}{2},0\right)$  and  $\left(-\frac{3\sqrt{5}}{2},0\right)$
- (C)  $(\sqrt{5},0)$  and  $(-\sqrt{5},0)$
5. The graph of the hyperbola  $\frac{x^2}{9} - \frac{y^2}{5} = 1$  is shown below.  $S$  is one of the foci and  $AB$  is a chord of the hyperbola that is perpendicular to the  $x$ -axis at  $S$ .



What is the length of  $AB$ ?

- (A)  $\frac{10}{3}$
- (B)  $\frac{18}{\sqrt{5}}$
- (C)  $\frac{10}{9}$
- (D)  $\frac{18}{5}$

**END OF SECTION I**

## Section II

Attempt all questions.

Answer each question in a separate writing booklet.

Your responses should include relevant mathematical reasoning and/or calculations.

---

### **Question 6** (16 marks)

(a) Evaluate  $\int_0^7 \frac{2}{\sqrt{x+9}} dx$ . 2

(b) Find  $\int \sin^3 x \, dx$  3

(c) (i) Write  $\frac{2x+3}{(x-2)(x^2+2)}$  in the form  $\frac{A}{x-2} + \frac{Bx+C}{x^2+2}$  2

(ii) Hence find  $\int \frac{2x+3}{(x-2)(x^2+2)} dx$  3

(d) Find  $\int \frac{dx}{\sqrt{13+4x-x^2}}$  3

(e) Find  $\int x \cos 2x \, dx$  3

**End of Question 6**

BLANK PAGE

**Question 7** (20 marks)

- (a) The point  $P(4\sec\theta, 3\tan\theta)$  lies on a hyperbola that has both vertices on the  $x$ -axis.
- (i) Find the equation of the hyperbola in Cartesian form. 1
- (ii) Find the eccentricity of the hyperbola. 1
- (iii) Draw a neat **HALF** page sketch of this hyperbola, showing the coordinates of the vertices and the foci, and the equations of the directrices and asymptotes. 3
- (iii) Derive the equation of the tangent to the hyperbola at the point  $P$  in parametric form. 2

- (b) The points  $P\left(2p, \frac{2}{p}\right)$  and  $Q\left(2q, \frac{2}{q}\right)$  lie on the rectangular hyperbola  $xy = 4$ .

$M$  is the midpoint of  $PQ$ .

$P$  and  $Q$  move on the hyperbola so that the chord  $PQ$  will always pass through the point  $R(4,2)$  and  $pq > 0$

- (i) Show that the chord  $PQ$  has the equation  $x + pqy = 2(p + q)$ . 2
- (ii) Show that  $pq = p + q - 2$ . 1
- (iii) Hence find the equation of the locus of  $M$  as  $P$  moves on the hyperbola. 3

**Question 7** (continued)

(c) (i) Using the substitution  $x = \tan \frac{1}{2} x$  or otherwise, show that  $\int_0^{\frac{\pi}{2}} \frac{1}{1 - \frac{1}{2} \sin x} dx = \frac{4\pi}{3\sqrt{3}}$  **3**

(ii) Show that  $\int_0^{2a} f(x) dx = \int_0^a [f(x) + f(2a - x)] dx$  **2**

(iii) Hence evaluate  $\int_0^{\pi} \frac{x}{1 - \frac{1}{2} \sin x} dx$  **2**

**END OF EXAMINATION**

BLANK PAGE



**SECTION I****Q1 D****Q2 A****Q3 D****Q4 C****Q5 A****SECTION II****Question 16****(a)**

$$\begin{aligned}\int_0^7 \frac{2}{\sqrt{x+9}} dx &= \int_0^7 2(x+9)^{-\frac{1}{2}} dx \\ &= \left[ 4(x+9)^{\frac{1}{2}} \right]_0^7 = \left[ 4\sqrt{x+9} \right]_0^7 \\ &= \left[ (4\sqrt{7+9}) - (4\sqrt{0+9}) \right] \\ &= 4\end{aligned}$$

**(b)**

$$\begin{aligned}\int \sin^3 x dx &= \int \sin x \sin^2 x dx \\ &= \int \sin x (1 - \cos^2 x) dx \\ &= \int \sin x - \sin x \cos^2 x dx \\ &= -\cos x + \frac{\cos^3 x}{3} + C\end{aligned}$$

**(c) (i)**

$$\begin{aligned}\frac{2x+3}{(x-2)(x^2+2)} &= \frac{A}{x-2} + \frac{Bx+C}{x^2+2} \\ 2x+3 &= A(x^2+2) + (Bx+C)(x-2)\end{aligned}$$

Sub  $x = 2$ 

$$7 = A(4+2)$$

$$A = \frac{7}{6}$$

Sub  $x = 0$ 

$$3 = A(0+2) + C(-2)$$

$$3 = \frac{7}{6}(2) + C(-2)$$

$$C = -\frac{1}{3}$$

Sub  $x = 1$ 

$$5 = A(1+2) + (B+C)(-1)$$

$$5 = 3\left(\frac{7}{6}\right) - B - \frac{1}{3}(-1)$$

$$B = \frac{7}{3} - 5 + \frac{1}{3}$$

$$B = -\frac{7}{6}$$

$$\frac{2x+3}{(x-2)(x^2+2)} = \frac{\frac{7}{6}}{x-2} + \frac{-\frac{7}{6}x - \frac{1}{3}}{x^2+2}$$

**(ii) ho**

$$\begin{aligned}\int \frac{2x+3}{(x-2)(x^2+2)} dx &= \int \frac{\frac{7}{6}}{x-2} dx + \int \frac{-\frac{7}{6}x - \frac{1}{3}}{x^2+2} dx \\ &= \frac{7}{6} \int \frac{1}{x-2} dx + \int \frac{-\frac{7}{6}x}{x^2+2} + \int \frac{-\frac{1}{3}}{x^2+2}\end{aligned}$$

$$\begin{aligned}&= \frac{7}{6} \int \frac{1}{x-2} dx - \frac{7}{12} \int \frac{2x}{x^2+2} dx - \frac{1}{3} \int \frac{1}{x^2+2} dx \\ &= \frac{7}{6} \ln(x-2) - \frac{7}{12} \ln(x^2+2) + \frac{1}{3\sqrt{2}} \tan^{-1} \frac{x}{\sqrt{2}} + C\end{aligned}$$

**(d)**

$$\begin{aligned}\int \frac{dx}{\sqrt{13+4x-x^2}} &= \int \frac{dx}{\sqrt{17-(x^2-4x+4)}} \\ &= \int \frac{dx}{\sqrt{17-(x-2)^2}} \\ &= \sin^{-1} \frac{x-2}{\sqrt{17}} + C\end{aligned}$$

**(e)**

$$\begin{aligned}\int x \cos 2x dx &= \frac{x \sin 2x}{2} - \int \frac{\sin 2x}{2} dx \\ &= \frac{x \sin 2x}{2} + \frac{\cos 2x}{4}\end{aligned}$$

**Question 7****a)****(i)** Find the equation of the hyperbola in Cartesian form.

$$(4 \sec \theta, 3 \tan \theta)$$

$$a = 4 \text{ and } b = 3$$

$$\text{Equation is } \frac{x^2}{16} - \frac{y^2}{9} = 1$$

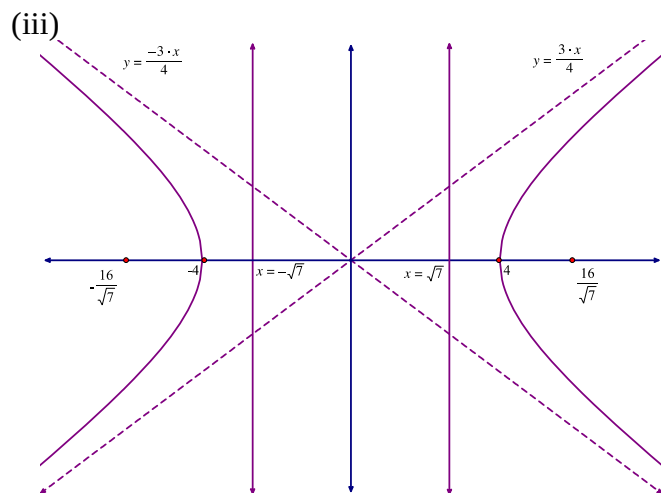
(ii) Find the eccentricity of the hyperbola.

$$b^2 = a^2(1 - e^2)$$

$$\frac{9}{16} = 1 - e^2$$

$$\frac{7}{16} = e^2$$

$$\frac{\sqrt{7}}{4} = e$$



(iv) Derive the equation of the tangent to the hyperbola at the point  $P$  in parametric form.

At  $P(4\sec\theta, 3\tan\theta)$

By Implicit Differentiation

$$\frac{2x}{16} - \frac{2y}{9} \times \frac{dy}{dx} = 0$$

$$\frac{2y}{9} \times \frac{dy}{dx} = \frac{2x}{16}$$

$$\frac{dy}{dx} = \frac{9x}{16y}$$

At  $P(4\sec\theta, 3\tan\theta)$

$$\frac{dy}{dx} = \frac{3\sec\theta}{4\tan\theta}$$

Equation of the tangent at  $P$

At  $P(4\sec\theta, 3\tan\theta)$

$$y - 3\tan\theta = \frac{3\sec\theta}{4\tan\theta}(x - 4\sec\theta)$$

$$4y\tan\theta - 12\tan^2\theta = 3x\sec\theta - 12\sec^2\theta$$

$$4y\tan\theta = 3x\sec\theta - 12\sec^2\theta + 12\tan^2\theta$$

$$4y\tan\theta = 3x\sec\theta + 12$$

b)

(i) Show that the chord  $PQ$  has the equation

$$x + pqy = 2(p + q)$$

Gradient of  $PQ$

$$m_{PQ} = \frac{\frac{2}{p} - \frac{2}{q}}{2p - 2q}$$

$$= \frac{2(q - p)}{2(p - q)}$$

$$= -\frac{1}{pq}$$

Equation of  $PQ$

$$y - \frac{2}{p} = -\frac{1}{pq}(x - 2p)$$

$$pqy - 2q = -x + 2p$$

$$x + pqy = 2(p + q)$$

(ii) Show that  $pq = p + q - 2$ .

At  $R(4, 2)$  Let  $x = 4$  and  $y = 2$

$$x + pqy = 2(p + q)$$

$$(4) + pq(2) = 2(p + q)$$

$$2 + pq = (p + q)$$

$$pq = (p + q) - 2$$

(iii) Hence find the equation of the locus of  $M$  as  $P$  moves on the hyperbola.

Midpoint of  $PQ$

$$x = \frac{2p + 2q}{2}$$

$$x = p + q$$

$$y = \frac{\frac{2}{p} + \frac{2}{q}}{2}$$

$$y = \frac{p + q}{pq}$$

Sub  $x = p + q$  and  $pq = (p + q) - 2$  into

$$y = \frac{p+q}{pq}$$

$$y = \frac{x}{(p+q)-2}$$

$$= \frac{x}{x-2}$$

But  $pq > 0$  so P and Q must be on the same branch and as PQ must pass through R then both p and q must be positive.

Thus the equation of the locus of M is

$$y = \frac{x}{x-2} \quad x > 2$$

### **Question 7** (continued)

(c) (i) Using the substitution  $x = \tan \frac{x}{2}$  or

otherwise, show that  $\int_0^{\frac{\pi}{2}} \frac{1}{1 - \frac{1}{2} \sin x} dx = \frac{4\pi}{3\sqrt{3}}$

$$\int_0^{\frac{\pi}{2}} \frac{1}{1 - \frac{1}{2} \sin x} dx = \int_0^1 \frac{1}{1 - \frac{1}{2} \times \frac{2t}{1+t^2}} \frac{2dt}{1+t^2} dx$$

$$= \int_0^1 \frac{2dt}{1+t^2-t}$$

$$= \int_0^1 \frac{2dt}{1 - \frac{1}{4} + t^2 - t + \frac{1}{4}}$$

$$= 2 \int_0^1 \frac{dt}{\frac{3}{4} + \left(t - \frac{1}{2}\right)^2}$$

$$= 2 \left[ \frac{2}{\sqrt{3}} \tan^{-1} \frac{2\left(t - \frac{1}{2}\right)}{\sqrt{3}} \right]_0^1$$

$$= \frac{4}{\sqrt{3}} \left[ \tan^{-1} \frac{2\left(\frac{1}{2}\right)}{\sqrt{3}} - \tan^{-1} \frac{2\left(-\frac{1}{2}\right)}{\sqrt{3}} \right]$$

$$= \frac{4}{\sqrt{3}} \left[ \frac{\pi}{6} - \left(-\frac{\pi}{6}\right) \right]$$

$$= \frac{4\pi}{3\sqrt{3}} \quad \quad \quad \mathbf{3}$$

(ii) Show that

$$\int_0^{2a} f(x) dx = \int_0^a [f(x) + f(2a-x)] dx$$

$$LHS = \int_0^{2a} f(x) dx$$

$$= \int_0^a f(x) dx + \int_a^{2a} f(x) dx$$

and

$$\int_a^{2a} f(x) dx = \int_0^a f(2a-x) dx \text{ by reflecting in the line } x = a$$

OR

$$\text{Let } u = 2a - x$$

$$du = -dx$$

$$\text{When } x = a \quad u = a$$

$$\text{When } x = 2a \quad u = 0$$

$$\int_a^{2a} f(u) du = \int_0^a f(2a-x) dx$$

$$LHS = \int_0^a f(x) dx + \int_a^{2a} f(x) dx$$

$$= RHS$$

$$\begin{aligned}
\int_0^{\pi} \frac{x}{1-\frac{1}{2}\sin x} dx &= \int_0^{\frac{\pi}{2}} \frac{x}{1-\frac{1}{2}\sin x} + \frac{\pi-x}{1-\frac{1}{2}\sin(\pi-x)} dx \\
&= \int_0^{\frac{\pi}{2}} \frac{x}{1-\frac{1}{2}\sin x} dx + \int_0^{\frac{\pi}{2}} \frac{\pi}{1-\frac{1}{2}\sin(\pi-x)} dx - \int_0^{\frac{\pi}{2}} \frac{x}{1-\frac{1}{2}\sin(\pi-x)} dx \\
&= \int_0^{\frac{\pi}{2}} \frac{x}{1-\frac{1}{2}\sin x} dx + \int_0^{\frac{\pi}{2}} \frac{\pi}{1-\frac{1}{2}\sin x} - \int_0^{\frac{\pi}{2}} \frac{x}{1-\frac{1}{2}\sin x} dx \\
&= \int_0^{\frac{\pi}{2}} \frac{\pi}{1-\frac{1}{2}\sin x} dx \\
&= \pi \int_0^{\frac{\pi}{2}} \frac{1}{1-\frac{1}{2}\sin x} dx = \frac{4\pi^2}{3\sqrt{3}}
\end{aligned}$$